[INCH-POUND]
A-A-59616B
29 September 2022
SUPERSEDING
A-A-59616A
27 February 2006

COMMERCIAL ITEM DESCRIPTION

PIPE FITTINGS: BUSHINGS, LOCKNUTS, AND PLUGS; IRON, STEEL, AND ALUMINUM; (THREADED); 125 - 150 POUNDS

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the general requirements for threaded pipe fittings: bushings, locknuts, and plugs made of iron, steel, and aluminum; 125 - 150 pounds. Threaded pipe fittings covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see 7.1).

Example of a plug PIN with left hand threads.

AA59616	A	B	Q	B	C	P	L
CID number	Type (see table I)	Class (see table I)	Style (see table I)	Material	Finish	Size	Blank for right hand thread L for left.

Example of a bushing PIN with right hand threads.

AA59616	B	E	Q	B	C	FD	
CID number	Type (see table I)	Class (see table I)	Style (see table I)	Material	Finish	Size 1/	Blank for right hand thread L for left.

1/ Bushings require two letters used to indicate the OD (outside diameter) and the ID (inside diameter).

2.1 Type, class, style and size. Plugs, bushings, and locknuts shall be of the following types, classes, styles, and sizes (see table I).

Beneficial comments, recommendations, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, Defense Supply Center Columbus, Attn: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or Email to: Construction@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 4730
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.

TABLE I. Type, class, style designators.

Type	Type designator	Class designator	Class characteristics	Style designator	Style characteristics	Applicable sizes (see 2.1.1)
Plug	A	A	Square head	C	Cored	D through M
				S	Solid	A through M
		B	Bar or slotted head	Q	N/A	N, P, R, and T
		C	Countersunk or headless	A	Hexagon socket	A through F
				B	Square socket	D through N
Bushing	B	D	Outside head	Q	N/A	Requires two letters to indicate the O.D. and the I.D. A through T
		E	Inside head			
		F	Face bushing			
Locknut 1/	C	Q	N/A	Q	N/A	A through T

1/ Material B only see 3.2.

2.1.1 Sizes. For size designators see table II.

TABLE II. Size designators.

Size designator	Fraction inch	Decimal Inch (mm)
A	1/8	.125 (3.18)
B	¼	.250 (6.35)
C	3/8	.375 (9.53)
D	½	.500 (12.70)
E	¾	.750 (19.05)
F	1	1.000 (25.40)
G	1 ¼	1.250 (31.75)
H	1 ½	1.500 (38.10)
J	2	2.000 (50.80)
K	2 ½	2.500 (63.50)
L	3	3.000 (76.20)
M	3 ½	3.500 (88.90)
N	4	4.000 (101.60)
P	5	5.000 (127.00)
R	6	6.000 (152.40)
T	8	8.000 (203.20)

NOTES:
1. Metric equivalents are given for information only.

2.2 Materials. Materials shall be as specified in 3.2.

2.3 Finishes. Finishes shall be as specified in 3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Pipe fittings: bushings, locknuts, and plugs; iron, steel, and aluminum; (threaded); 125 - 150 pounds supplied to this CID shall be as specified herein.

3.2 Materials. Materials shall be as specified in table III.

TABLE III. Materials.

Material designator	Material	Reference document
A	Aluminum	ASTM B211, Alloy 1100, 2011, or 2024
B	Iron or Steel	ASME B16.14 (iron or steel) 1/

1/ Class C steel plugs from Table I, 1 inch or smaller, which do not require cadmium plating may be in accordance with SAE J531 (NPTF only).

TABLE IV. Finishes.

Fitting Material	Finish designator	Finish	Reference document
Aluminum	A	Anodized	ASTM B580
	N	Non-anodized	N/A
	W	Anodized (without hexavalent chromium)	MIL-PRF-8625
Iron or steel	C	Electrodeposited zinc zinc (without hexavalent chromium) 1/	ASTM B633
	D	Electrodeposited cadmium 1/ 2/	ASTM B766
	U	Uncoated	N/A
	Z	Electrodeposited zinc-nickel 1/	SAE AMS2461

1/ Coating shall be applied prior to all threading.
2/ Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable substance list.

3.4 Threads. Threads shall be in accordance with ASME B1.20.1. Pipe plugs manufactured to SAE J531 (NPTF) shall be in accordance with ASME B1.20.3.

3.4.1 Bushings. Class D bushings shall have the external threads oversize in accordance with ASME B16.14.

3.4.2 Locknuts. All locknuts shall have internal threads with American National Standard straight pipe threads for locknut connections (NPSL) in accordance with ASME B1.20.1.

3.5 Performance. Bushings and plugs shall be capable of withstanding an air pressure of at least 60 pounds per square inch (psi) (4.14 bar) while under water without leaking at any part of the surface. Bushings and plugs shall also be capable of withstanding a hydrostatic pressure of either 250 psi (17.24 bar), for 125-pound (56.70 kg) fittings, or 300 psi (20.68 bar), for 150-pound (68.04 kg) fittings, for 1 minute without leaking or sweating.

3.6 Workmanship. Bushings, locknuts, and plugs supplied to this CID shall be clean, free of sand and scale, and be sound, smoothly cored when required, and true to form. There shall be no cold shuts, blowholes, porosity, or any other defects that may affect serviceability.

3.7 Marking. Threaded pipe fittings supplied to this CID shall be marked with the manufacturers (MFR's) standard commercial PIN. The part number on the unit pack shall be the CID PIN.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.1 Ozone Depleting Substances (ODS). Class I and II ozone depleting substances (ODS) shall not be used in A-A-59616 or any referenced procedures.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

7.2 Commercial and Government Entity (CAGE) code. CAGE code 58536 should be used for ordering purposes, inventory control, and submission of these bushings, locknuts, and plugs to Defense Supply Center Columbus (DSCC), aka Land and Maritime, under the Military Parts Control Advisory Group (MPCAG) evaluation program.

7.3 Source of documents.

Federal Regulations

FEDERAL ACQUISITION REGULATION (FAR)

- Federal Acquisition Regulations

(Copies of this document are available online at https://www.acquisition.gov/.)

MILITARY PERFORMANCE SPECIFICATIONS

MIL-PRF-8625 - Anodic Coatings for Aluminum and Aluminum Alloy

(Copies of these documents are available online at https://assist.dla.mil.)

Other Publications

ASME INTERNATIONAL

ASME B1.20.1	-	Pipe Threads, General Purpose (Inch)
ASME B1.20.3	-	Dryseal Pipe Threads (Inch)
ASME B16.14	-	Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.

(Copies of these documents are available online at https://www.asme.org/.)

ASTM INTERNATIONAL

ASTM B211	-	Aluminum and Aluminum-Alloy Bar, Rod, and Wire
ASTM B580	-	Anodic Oxide Coatings on Aluminum
ASTM B633	-	Electrodeposited Coatings of Zinc on Iron and Steel
ASTM B766	-	Electrodeposited Coatings of Cadmium

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

SAE J531	-	Automotive Pipe, Filler, and Drain Plugs
SAE AMS2461	-	Plating, Zinc-Nickel Alloy (12 to 16% Ni)

(Copies of this document are available online at https://www.sae.org/.)

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

d. When left-hand threads are required (see 3.4).

c. Packaging requirements.

7.5 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's Name and Address
12502	General Plug and Manufacturing Corporation 455 Main Street Grafton, OH 44044 800-289-7584
None	Billco Corporation 770 W. Belden Avenue Addison, IL 60101 630-543-4777 www.billcocorp.com
80996	Anvil International (Formerly ITT Grinnell Corporation) 110 Corporate Drive, Suite 10 P.O. Box 3180 Portsmouth, NH 03802-3180 603-422-8000 www.anvilintl.com

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.

7.6 Part number (P/N) supersession data. These CID PINs supersede the following MFG'S P/N's as shown in table V. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE V. P/N data.

Dash Number		
(See section 2)	MFR's	MFR's
A-A-59616	CAGE	P/N 1/
AA59616BDQBC	80996	383BLK3/8X1/4
AA59616BDQBUJH	80996	383BLK2X1-1/2
AA59616AASBD	80996	388BLK2
AA59616AASBOG	45152	2AF27S

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown. For actual part marking requirements see 3.7.

7.7 Supersession data. A-A-59616 supersedes WW-P-471.

7.8 Government users. To acquire information on obtaining these threaded pipe fittings: bushings, locknuts, and plugs; made of iron, steel, and aluminum; 125 - 150 pounds from the Government inventory system, please contact Defense Supply Center, Columbus, ATTN: DSCC Call Center (DSCC-NAB), P.O. Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-2271 or -3191.

7.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Review activities:
Army - AR, EA
Navy - CG, MC, SA

CIVIL AGENCY COORDINATING ACTIVITY:

GSA/FSS

Preparing activity:
DLA - CC

(Project 4730-2022-026)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:21Z
Check the source to verify that this is the current version before use.